

# CNC TRAINERS



# CNC TRAINERS

## DICT0200 - CNC LATHE TRAINER

### Controller :-

|                                |                                                |
|--------------------------------|------------------------------------------------|
| Type of Control System         | : PLC based System                             |
| Main Spindle Drive             | : AC Motor with VFD                            |
| Z and X Drives                 | : Stepper Drive with Ramp function             |
| Input to Stepper               | : Optically Isolated                           |
| Motor Drive                    | : TTL Compatible                               |
| Run Speed Control              | : Computer controlled with Software            |
| Interpolation                  | : Linear / Circular / Parabolic                |
| Cycles Used                    | : G, M, S, T, F, I, K, M, F                    |
| Cycles                         | : Circular Interpolation / Canned              |
| Threading                      | : Straight / Taper                             |
| Power Devices                  | : Short Circuit and over Temperature protected |
| Real Time Tool Path Simulation | : Provided                                     |

### SPINDLE

|               |                       |
|---------------|-----------------------|
| Spindle Speed | : Continuous Variable |
| Spindle Power | : 2 HP                |
| Inside Taper  | : Mt3                 |
| Speed Range   | : 50 to 3000 RPM      |

### Machine Details :-

|                         |                     |
|-------------------------|---------------------|
| Center Height           | : 100mm             |
| Swing Over Carriage     | : 80mm              |
| Distance between Center | : 300mm             |
| Longitudinal Travel     | : 310mm             |
| Weight                  | : 800kg             |
| Dimension in mm         | : 1300 x 800 x 1400 |
| Rapid Travel            | : 700 mm/min        |
| Resolution Z Axis       | : 5 micron          |
| Resolution X Axis       | : 5 micron (Dia.)   |
| Automatic               | : Provided          |

### AXIS TRAVEL

|        |          |
|--------|----------|
| X Axis | : 80 mm  |
| Z Axis | : 310 mm |



## DICT0201 - CNC LATHE TRAINER - SLANT BED

### Controller :-

|                               |                                                                         |
|-------------------------------|-------------------------------------------------------------------------|
| Type of control system        | : PLC based system                                                      |
| Axis motor and drive          | : Servo motor & drive with feedback                                     |
| Input to Servo                | : Optically Isolated                                                    |
| Motor Drive                   | : TTL compatible                                                        |
| Interpolation                 | : Linear / Circular / Parabolic                                         |
| Cycles Used                   | : G, M, S, T, F, I, K, M, F                                             |
| Cycles                        | : Circular Interpolation / Canned                                       |
| Threading                     | : Straight / Taper                                                      |
| Power Devices                 | : Short Circuit and over Temperature protected, Single phase protection |
| Real Time ToolPath Simulation | : Provided                                                              |

### SPINDLE

|               |                       |
|---------------|-----------------------|
| Spindle Speed | : Continuous Variable |
| Spindle Power | : 2 HP                |
| Inside Taper  | : Mt3                 |
| Speed Range   | : 100 to 3000 RPM     |

### Machine Details :-

|                         |                        |
|-------------------------|------------------------|
| Center Height           | : 100mm                |
| Swing Over Carriage     | : 80mm                 |
| Distance between Center | : 825mm                |
| Weight                  | : 700kg                |
| Dimension in mm         | : 1250 x 900 x 1500    |
| Rapid Travel            | : 700 mm/min           |
| Resolution Z Axis       | : 5 micron             |
| Resolution X Axis       | : 5 micron (Dia.)      |
| Automatic Lubrication   | : Provided             |
| Tool Charger            | : Automatic - 8 Micron |

### AXIS TRAVEL

|        |          |
|--------|----------|
| X Axis | : 100 mm |
| Z Axis | : 300 mm |



# CNC TRAINERS

## DICT0202 - CNC LATHE TRAINER WITH SERVO DRIVES

### Controller :-

|                                |                                                |
|--------------------------------|------------------------------------------------|
| Type of Control System         | : PLC based System                             |
| Main Spindle Drive             | : AC Motor with VFD                            |
| Z and X Drives                 | : Stepper Drive with Ramp function             |
| Input to Stepper Motor         | : Optically Isolated                           |
| Motor Drive                    | : TTL Compatible                               |
| Run Speed Control              | : Computer controlled with Software            |
| Interpolation                  | : Linear / Circular / Parabolic                |
| Cycles Used                    | : G, M, S, T, F, I, K, M, F                    |
| Cycles                         | : Circular Interpolation / Canned              |
| Threading                      | : Straight / Taper                             |
| Power Devices                  | : Short Circuit and over Temperature protected |
| Real Time Tool Path Simulation | : Provided                                     |

### SPINDLE

|               |                       |
|---------------|-----------------------|
| Spindle Speed | : Continuous Variable |
| Spindle Power | : 2 HP                |
| Inside Taper  | : Mt3                 |
| Speed Range   | : 50 to 3000 RPM      |

### Machine Details :-

|                         |                     |
|-------------------------|---------------------|
| Center Height           | : 100mm             |
| Swing Over Carriage     | : 80mm              |
| Distance between Center | : 250mm             |
| Longitudinal Travel     | : 250mm             |
| Weight                  | : 350kg             |
| Dimension               | : 1100 x 750 x 1700 |
| Rapid Travel            | : 500 mm/min        |
| Resolution X & Z Axis   | : 10 micron         |
| Dt. Motor Power         | : 40 - 4A           |
| Lubrication             | : Manual            |



## DICT0203 - CNC LATHE TRAINER

### Controller :-

|                                |                                                                         |
|--------------------------------|-------------------------------------------------------------------------|
| Type of control system         | : PLC based system                                                      |
| Axis motor and drive           | : Servo motor & drive with feedback                                     |
| Input to Servo                 | : Optically Isolated                                                    |
| Motor Drive                    | : TTL compatible                                                        |
| Interpolation                  | : Linear / Circular / Parabolic                                         |
| Cycles Used                    | : G, M, S, T, F, I, K, M, F                                             |
| Cycles                         | : Circular Interpolation / Canned                                       |
| Threading                      | : Straight / Taper                                                      |
| Power Devices                  | : Short Circuit and over Temperature protected, Single phase protection |
| Real Time Tool Path Simulation | : Provided                                                              |

### SPINDLE

|               |                       |
|---------------|-----------------------|
| Spindle Speed | : Continuous Variable |
| Spindle Power | : 1.5 HP              |
| Inside Taper  | : Mt3                 |
| Speed Range   | : 100 to 3000 RPM     |

### Machine Details :-

|                         |                     |
|-------------------------|---------------------|
| Center Height           | : 100mm             |
| Swing Over Carriage     | : 80mm              |
| Distance between Center | : 250mm             |
| Longitudinal Travel     | : 250mm             |
| Weight                  | : 350kg             |
| Dimension               | : 1100 x 750 x 1700 |
| Rapid Travel            | : 500 mm/min        |
| Resolution X & Z Axis   | : 10 micron         |
| Dt. Motor Power         | : 40V - 4A          |
| Lubrication             | : Manual            |



# CNC TRAINERS

## DICT0204 - CNC LATHE TRAINER

### Controller :-

|                                   |                                                                  |
|-----------------------------------|------------------------------------------------------------------|
| Type of Control System            | : Siemens / Fanuc                                                |
| Main Spindle Drive                | : AC Motor with VFD                                              |
| Z and X Drives                    | : Servo Drive with Servo Motor                                   |
| Input to Servo                    | : Optically Isolated                                             |
| Motor Drive                       | : Smart & Intelligent                                            |
| Run Speed Control                 | : Through controller                                             |
| Interpolation                     | : Linear / Circular / Parabolic                                  |
| Cycles Used                       | : G, M, S, T, F, I, K, M, F<br>: Circular Interpolation / Canned |
| Threading                         | : Straight / Taper                                               |
| Power Devices                     | : Short Circuit and over<br>: Temperature protected              |
| Real Time Tool Path<br>Simulation | : Provided                                                       |

### Machine Details :-

|                         |                         |
|-------------------------|-------------------------|
| Center Height           | : 150mm                 |
| Swing Over Carriage     | : 125mm                 |
| Distance between Center | : 825mm                 |
| Weight                  | : 1100mm                |
| Dimension in mm         | : 2000 x 1200 x 1700    |
| Rapid Travel            | : 5000 mm/min           |
| Resolution Z Axis       | : 1 micron              |
| Resolution X Axis       | : 1 micron (Dia.)       |
| Automatic Lubrication   | : Provided              |
| Tool Charger            | : Automatic - 4 station |

### SPINDLE

|               |                       |
|---------------|-----------------------|
| Spindle Speed | : Continuous Variable |
| Spindle Power | : 3 HP                |
| Inside Taper  | : Mt4                 |
| Speed Range   | : 50 to 1800 RPM      |

## DICT0205 - CNC MILL TRAINER

### Controller :-

|                          |                                                             |
|--------------------------|-------------------------------------------------------------|
| Controller               | : SIEMENS 8020                                              |
| Main Spindle Drive       | : D. C. Thyristor Drive, Inbuilt Safety,<br>1.5 HP          |
| X,Y and Z axes drive     | : SIEMENS servo motors and Drives                           |
| Input to Stepper         | : Optically Isolated,                                       |
| Motor Drive TTL          | : Linear, Circular Parabolic                                |
| Compatible Interpolation | : interpolation, Slot / Pocket Milling                      |
| Code Used                | : G, M, S, T, F, I, K, M, F                                 |
| Cycles                   | : Circular Interpolation,<br>: Mirror, Pocket Milling Cycle |
| Pocket & Helical Milling | : Provided with Std Codes                                   |

### Table :-

|                |                       |
|----------------|-----------------------|
| Length X Width | : 450 x 175 (Approx.) |
| Clamping Area  | : 350 x 130 (Approx.) |
| T Slots        | : 10mm slots width    |

### Machine Details :-

|                       |                        |
|-----------------------|------------------------|
| Resolution            | : 5 micron             |
| Tool Change           | : Automatic, 8 Station |
| Rapid Travel          | : 5000 mm / min        |
| Weight in kg          | : 750 kg.              |
| Dimension in mm       | : 850 x 850 x 1550mm   |
| Automatic Lubrication | : Provided             |
| Repeatability         | : 10 micron            |



### SPINDLE

|               |                       |
|---------------|-----------------------|
| Spindle Speed | : Continuous Variable |
| Spindle Power | : 1.5 HP              |
| Inside Taper  | : ISO 30              |
| Speed Range   | : 100 to 4000 RPM     |

### AXIS TRAVEL

|                       |         |
|-----------------------|---------|
| Longitudinal Traverse | : 250mm |
| Cross Traverse        | : 175mm |
| Vertical Traverse     | : 250mm |

# CNC TRAINERS

## DICT0206 - CNC MILL TRAINER

### Controller :-

|                          |                                                              |
|--------------------------|--------------------------------------------------------------|
| Type of Control System   | : PLC based system Under collaboration from                  |
| Main Spindle Drive       | : D. C. Thyristor Drive, Inbuilt Safety                      |
| X, Y and Z axes drive    | : Stepper Motor Drive Servo Drive                            |
| Input to Stepper         | : Optional<br>Optically Isolated, Motor Drive TTL compatible |
| Run Speed Control        | : Computer Control with Software                             |
| Interpolation            | : Linear, Circular Parabolic interpolation                   |
| Code Used                | : G, M, S, T, F, I, K, M, F                                  |
| Cycles                   | : Circular Interpolation, Mirror, Pocket Milling Cycle       |
| Pocket & Helical Milling | : Provided with Std Codes                                    |
| Real Time Tool Path      | : With online 3D simulation                                  |

### Table :-

|                |                       |
|----------------|-----------------------|
| Length X Width | : 600 x 225 (Approx.) |
| Clamping Area  | : 450 x 160 (Approx.) |
| T Slots        | : 10mm slots width    |

### SPINDLE

|               |                       |
|---------------|-----------------------|
| Spindle Speed | : Continuous Variable |
| Spindle Power | : 1.5 HP              |
| Inside Taper  | : ISO 30              |
| Speed Range   | : 300 to 3000 RPM     |

### Machine Details :-

|                       |                                    |
|-----------------------|------------------------------------|
| Resolution            | : 5 micron                         |
| Distance From Spindle | : 25 - 225 mm ( Nose to Table Top) |
| Centerline to Column  | : 1700 mm                          |
| Rapid Travel          | : 700mm / min                      |
| Weight in kg          | : 950 kg.                          |
| Dimension in mm       | : 1200 x 1200 x 1900mm             |



## DICT0207 - CNC LATHE TRAINER

### Controller :-

|                          |                                                                                     |
|--------------------------|-------------------------------------------------------------------------------------|
| Type of Control System   | : PLC based system                                                                  |
| Main Spindle Drive       | : D. C. Thyristor Drive, Inbuilt Safety                                             |
| X, Y and Z axis Drives   | : Stepper Motor Drive                                                               |
| Input to Stepper         | : Optically Isolated,                                                               |
| Motor Drive              | : TTL compatible                                                                    |
| Run Speed Control        | : Computer Control with Software                                                    |
| Interpolation            | : Linear, Circular, Parabolic                                                       |
| Cycles Used              | : Interpolation                                                                     |
| Cycles                   | : G, M, S, T, F, I, K, M, F<br>Circular Interpolation, Mirror, Pocket Milling Cycle |
| Pocket & Helical Milling | : Provided                                                                          |
| Real Time Tool Path      | : Provided simulation                                                               |

### Table :-

|                |                       |
|----------------|-----------------------|
| Length X Width | : 450 x 175 (Approx.) |
| Clamping Area  | : 360 x 130 (Approx.) |
| T Slots        | : 10mm slots width    |

### Machine Details :-

|                       |                         |
|-----------------------|-------------------------|
| Resolution            | : 5 micron              |
| Distance From Spindle | : 700mm / min           |
| Tool Change           | : Quick, Change, Manual |
| Centerline to Column  | : 140 mm                |
| Weight in kg          | : 550 kg.               |
| Dimension in mm       | : 850 x 800 x 1630mm    |

### SPINDLE

|               |                       |
|---------------|-----------------------|
| Spindle Speed | : Continuous Variable |
| Spindle Power | : 1 HP                |
| Inside Taper  | : ISO 30              |
| Speed Range   | : 300 to 3000 RPM     |